CampusNotes - Academic Study Resource

DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING - SEMESTER 5

OFFICIAL ACADEMIC STUDY MATERIAL - CSE CURRICULUM

Experiment 8: Query Execution Plan Tuning with EXPLAIN ANALYZE

Course: Advanced Database Engineering Laboratory [23CS3506]
Category: Experiments / Practicals | Experiment #8 | Academic Year: 2026-27

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1. Overview & Objective:

Interpret query plans in PostgreSQL, identifying cost bottlenecks, nested loop joins, and hash joins.

2. Practical Tasks & Verification Steps:

1. Run EXPLAIN ANALYZE on unindexed 3-table join query and identify cost and actual time metrics.
2. Analyze Seq Scan vs Index Scan vs Bitmap Index Scan plan nodes.
3. Create appropriate composite index student_id, semester and observe plan change to Index Scan.
4. Compare execution latency and buffer hits before and after index creation.

Submission Guidelines & Verification:

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